TAPASH KUMAR

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SUMMARY

Data Scientist specializing in Machine Learning and Deep Learning to solve complex analytical challenges. Proficient in building predictive models with TensorFlow and Keras, performing advanced data modelling with SQL, and delivering actionable business insights through Power BI. Experienced in deploying end-to-end AI solutions that combine technical rigor with practical scalability to drive informed decision-making.

TECHNICAL SKILLS

- Programming & Databases: Python, Microsoft Server SQL, PostgreSQL, TypeScript
- Data Analysis & Statistics: Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering
- Machine Learning: Supervised and Unsupervised Learning, Hyperparameter Tuning, Model Evaluation
- Deep Learning: Convolutional Neural Network (CNN), Natural Language Processing (NLP)
- Libraries & Frameworks: NumPy, Pandas, scikit-learn, TensorFlow, Keras
- Generative AI: Large Language Models (LLMs), LangChain, Hugging Face Transformers, Retrieval-Augmented Generation (RAG)
- Vector Search & Retrieval: Vector Embeddings, FAISS, Chroma, Pinecone, Semantic Search
- Visualization & BI Tools: Matplotlib, Seaborn, Power BI, Excel
- Deployment, MLOps & Cloud: Flask, FastAPI, Streamlit, Docker, CI/CD, Git/GitHub, Jupyter, Microsoft Azure

EXPERIENCE

InfoZeb Consultancy - AI Engineer

Gurugram — Mar 2026 – Present

- Worked on an AI-driven HR automation system to streamline recruitment workflows and decision-making.
- Implemented LLM-based candidate screening and ranking, along with intelligent CTA logic for automated workflow actions.

Stabor India Private Limited - AI Engineering Intern

Gurugram — Dec 2025 – Feb 2026

- Built and optimized a XGBoost model to improve demand forecasting accuracy by 15–20% on real-time datasets.
- Designed time-series data pipelines and tuned L1/L2 regularization to reduce overfitting and ensure robust model performance.

CSIR – National Physical Laboratory (NPL) - Intern

New Delhi — Jun 2025 – Jul 2025

- Applied AI/ML-driven data analysis to compute Allan Deviation, model atomic clock noise, and benchmark drift against Bureau International des Poids et Mesures standards.

ISRO Telemetry, Tracking and Command Network (ISTRAC) - Intern

Bangalore — Jun 2024 – Jul 2024

- Leveraged AI-assisted simulation and data analysis to optimize microwave cavity design in Ansys for TE011 mode, achieving ~44 Hz precision and improving Hydrogen Maser stability.

PROJECTS

NLP-Based Movie Recommendation System — Demo

Tech Stack: LangChain, FAISS, HuggingFace, FastAPI, PostgreSQL, Docker, Next.js, Firebase Auth

- Built **FastAPI** recommender generating **HuggingFace embeddings via LangChain** and retrieving similar movies using FAISS indexing.
- Processed **5K+ movie** metadata vectors enabling semantic similarity search across plots, genres, and keywords.
- Achieved sub-second recommendations by optimizing **FAISS vector index** and embedding inference pipeline.
- Deployed full-stack system with **Next.js, PostgreSQL, Firebase Auth, and Docker** enabling scalable personalized recommendations.

Brain MRI Prediction Using CNN

Tech Stack: Python, Scikit-learn, Tensorflow, Keras, Matplotlib, Seaborn

- Engineered a custom **TensorFlow/Keras CNN** for brain tumor detection using data augmentation and hyperparameter tuning.
- Increased **model accuracy by 12%** over the baseline through rigorous image preprocessing and architectural refinement.
- Evaluated model performance metrics using scikit-learn, maintaining a **94% F1-score** to balance sensitivity and specificity.

AI Intelligent Chatbot

Tech Stack: LangGraph, LangChain, Streamlit, SQLite

- Built stateful conversational chatbot using LangGraph checkpoints to maintain persistent memory across sessions.
- Implemented LangChain orchestration for prompt management and automated conversation titling.
- Developed Streamlit interface with SQLite persistence for chat history storage and session continuity.

CERTIFICATES

- Google Data Analytics Professional Certificate
- IBM AI Enterprise Workflow
- Pregrad Machine Learning Course
- IIRS AI/ML for Geodata Analysis

EDUCATION

Delhi Technological University (DTU) | **B.Tech. Engineering Physics** | Oct 2022 – May 2026

ACHIEVEMENTS

- Published research paper **"Innovative Cavity Design for Hydrogen Masers Using Simulation Methods"** in ISJEM (Mar 2026).